PAVITHRA D

Mechanical Engineering Intern Candidate — Manufacturing & Production

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PROFESSIONAL SUMMARY

Final-year Mechanical Engineering student with two prototypes shipped end-to-end — an IoT-enabled waste-to-filament extrusion system and an ESP32-based EV regenerative braking system — applying DFM and GD&T practice across both builds. IEEE RAS Vice Chair coordinating cross-functional teams and industry workshops. Seeking a Manufacturing & Production internship to bring the same build-fast, iterate-faster approach to a live production floor.

EDUCATION

Vemana Institute of Technology (VTU) — B.E. Mechanical Engineering, CGPA 7.9/10 — 09/2023 – 06/2027 | Karnataka, India

Krupanidhi Group of Institutions — KSEAB, 68.8% — 2021 – 2023 | Karnataka, India

Patel Public School — ICSE, KSEEB, 71.2% — 2021 | Karnataka, India

TECHNICAL SKILLS

- **Design & CAD:** AutoCAD, Fusion 360 (incl. additive manufacturing workflows), GD&T fundamentals, DFM/DFMA principles
- **Simulation & Analysis:** Ansys, Finite Element Analysis, MATLAB Simulink
- **Manufacturing & Production:** Basic CNC Programming, 3D Printing/Additive Manufacturing, Lean Manufacturing & 5S fundamentals, BOM preparation, Total Quality Management (TQM) coursework
- **Programming & Tools:** Python, MS Office, LaTeX

PROJECTS

IoT-Based Automated 3D Printer Filament Extrusion System Using Solid Waste

- Cut material waste and fixture lead time — measured by fewer rework cycles against a manual extrusion baseline — by designing and building an IoT-enabled system converting plastic waste into 3D-printer filament.
- Applied DFM principles and GD&T tolerancing while maintaining an accurate BOM across design iterations, using Fusion 360, Arduino/ESP32, 3D printing, and CNC machining.

Adaptive Regenerative Braking System

- Improved braking efficiency, energy recovery, and vehicle stability against a fixed-braking baseline by designing an adaptive system that adjusts braking intensity to road gradient using an ESP32 microcontroller and ADXL335 sensor.
- Resolved intermittent braking lag by conducting root cause analysis on sensor calibration error and recalibrating the accelerometer input pipeline — built with BLDC motor, PWM control, and Embedded C.

LEADERSHIP & INVOLVEMENT

- Vice Chair, IEEE Robotics and Automation Society (RAS) Student Chapter (2025–2026) — organized Industrial Robotics Skill Development Workshops and technical events; coordinated with faculty and industry professionals.
- Treasurer, IEEE RAS Student Chapter (2023–2024) — managed chapter finances and budget records; coordinated event expenditures for technical workshops and competitions.
- 2nd Prize, Robotics Competition (2025) — organized by Robomanthan Pvt. Ltd.

CERTIFICATIONS

- **Core:** Fusion 360 Certificate of Completion • Autodesk Certificate of Completion • MATLAB Onramp Certificate • Data Visualisation: Empowering Business with Effective Insights (Tata x Forage, 2026)
- **Additional:** Industrial Robotics Workshop (3-Day) • Fundamentals of Robotics (2-Day) • EV Saksham EV Workshop (5-Day) • Python 3.4.3

LANGUAGES

English (Professional) • Kannada (Conversational) • Hindi (Conversational) • Telugu (Native)